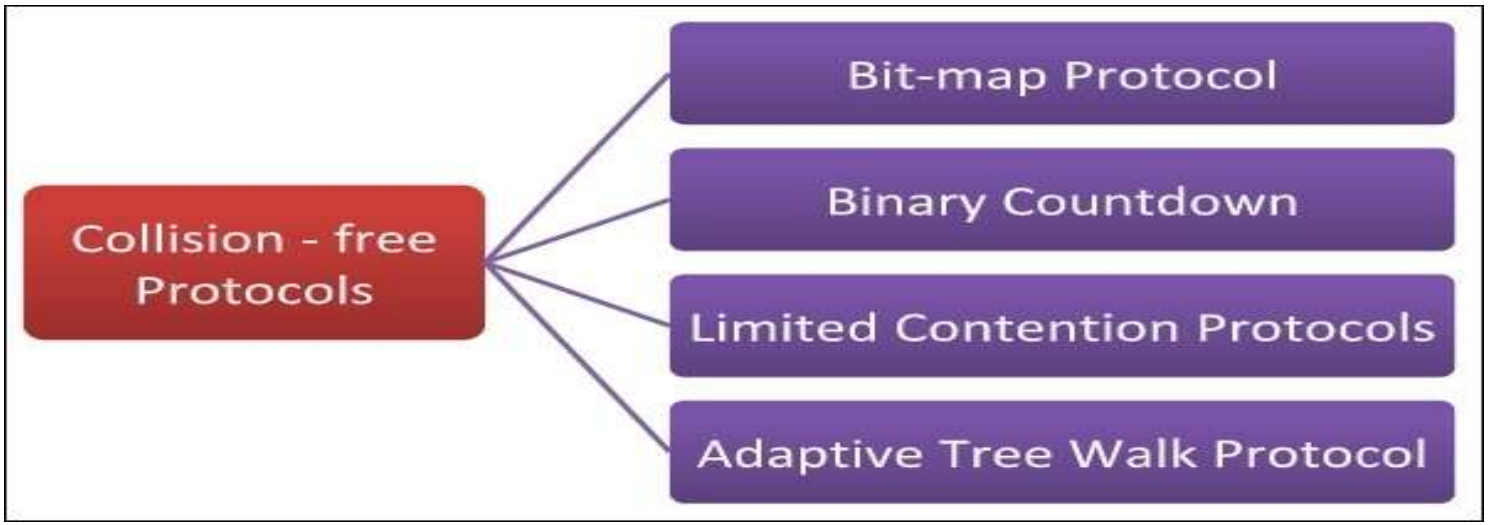
Collision Free Protocols:

- Pure and slotted Aloha, CSMA and CSMA/CD are **Contention based Protocols:**
- Try-if collide-Retry
  - No guarantee of performance
    - What happen if the network load is high?
- **Collision Free Protocols:**
  - Pay constant overhead to achieve performance guarantee
  - Good when network load is high

Collision-Free Protocols

- Almost collisions can be avoided in **CSMA/CD**.
- They can still occur during the contention period.
- The collision during contention period adversely affects the system performance,
- This happens when the cable is long and length of packet are short.
- This problem becomes serious as fiber optics network come into use. Here we shall discuss some protocols that resolve the collision during the contention period.
- Not only do collisions reduce bandwidth, but they make the time to send a frame variable, which is not a good fit for real-time traffic

Collision-Free Protocols



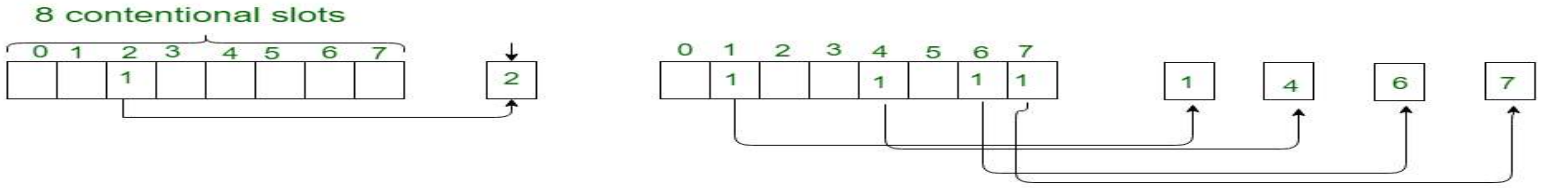
Collision-Free Protocols

Protocols that resolve the contention for the channel without any collisions at all, not even during the contention period.

- we assume that there are exactly N stations, each programmed with a unique address from 0 to N − 1.
- It does not matter that some stations may be inactive part of the time.
- We also assume that propagation delay is negligible

Bit-map Protocol:

- In In bitmap protocol method, each contention period consists of exactly N slots.
- If any station has to send frame, then it transmits a 1 bit in the respective slot.
- For example if station 2 has a frame to send, it transmits a 1 bit during the second slot.



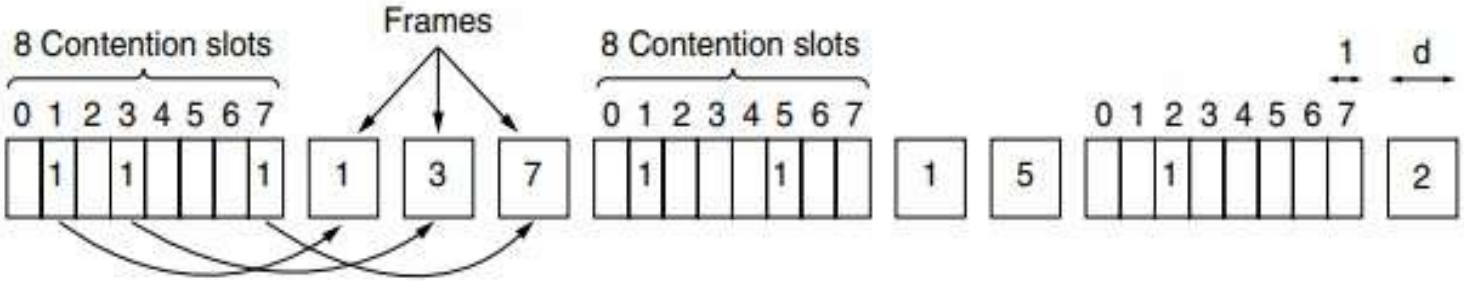
A Bit-map Protocol.

Bit-map Protocol:

- No other station is allowed to transmit during this slot.
- Regardless of what station 0 does, station 1 gets the opportunity to transmit a 1 bit during slot 1, but only if it has a frame queued.
- In general, station j may announce that it has a frame to send by inserting a 1 bit into slot j.
- After all N slots have passed by, each station has complete knowledge of which stations wish to transmit. At that point, they begin transmitting frames in numerical order
- There will never be any collisions because everyone agrees on who goes next.
- Protocols like this in which **the desire to transmit is broadcasting for the actual transmission are called *Reservation Protocols*.**

Bit-map Protocol:

- Since everyone agrees on who goes next, there will never be any collisions.
- After the last ready station has transmitted its frame, an event all stations can easily monitor, another N-bit contention period is begun.
- If a station becomes ready just after its bit slot has passed by, it is out of luck and must remain silent until every station has had a chance and the bit map has come around again.



Performance of **Bit-map Protocol**

- For analyzing the performance of this protocol, We will measure time in units of the contention bits slot, with a data frame consisting of *d* time units.
- Under **low load condition**s, the bitmap will simply be repeated over and over, for lack of data frames.
- Consider the situation from the point of view of a **low-numbered station**, such as 0 or 1.
  - Typically, when it becomes ready to send, the ‘‘current’’ slot will be somewhere in the middle of the bit map.
  - On average, the station will have to wait N /2 slots for the current scan to finish and another full N slots for the following scan to run to completion before it may begin transmitting
- The prospects for **high-numbered stations** are brighter. Generally, these will only have to wait half a scan (N /2 bit slots) before starting to transmit
  - High numbered stations rarely have to wait for the next scan.
- Since low-numbered stations must wait on average 1.5N slots and high-numbered stations must wait on average 0.5N slots, the mean for all stations is N slots.
- The overhead per frame is N bits and the amount of data is d bits,
- for an efficiency of d /(d + N)

Performance of **Bit-map Protocol**

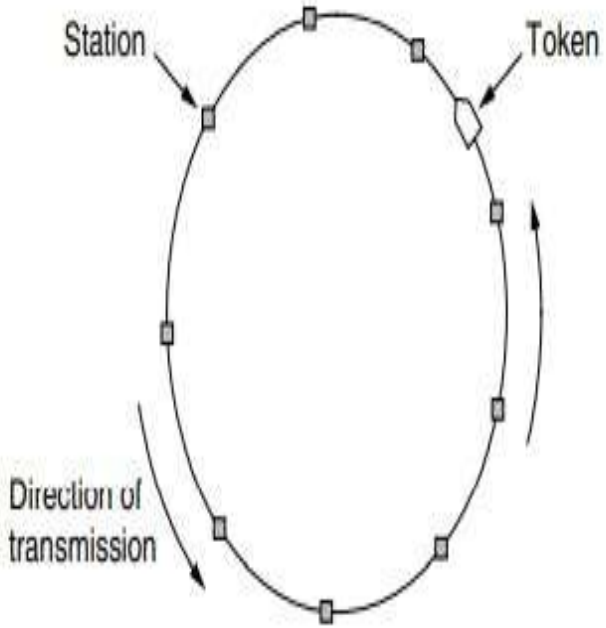
- All the stations have something to send all the time at high load, the N bit contention period is prorated over N frames, yielding an overhead of only 1 bit per frame.
- or an efficiency of d /(d + 1)
- The mean delay for a frame is equal to the sum of the time it queues inside its station, plus an additional (N − 1)d + N once it gets to the head of its internal queue.
- This interval is how long it takes to wait for all other stations to have their turn sending a frame and another bitmap.

Token Passing

- The essence of the bit-map protocol is that it lets every station transmit a frame in turn in a predefined order.
- Another way to accomplish the same thing is to pass a small message called a token from one station to the next in the same predefined order.
- The token represents permission to send. If a station has a frame queued for transmission when it receives the token, it can send that frame before it passes the token to the next station.
- If it has no queued frame, it simply passes the token.

Token ring protocol

- In a token ring protocol, the topology of the network is used to define the order in which stations send.
- The stations are connected one to the next in a single ring.
- Passing the token to the next station then simply consists of receiving the token in from one direction and transmitting it out in the other direction
- Frames are also transmitted in the direction of the token. This way they will circulate around the ring and reach whichever station is the destination.
- However, to stop the frame circulating indefinitely (like the token), some station needs to remove it from the ring.
- This station may be either the one that originally sent the frame, after it has gone through a complete cycle, or the station that was the intended recipient of the frame.



Token ring protocol

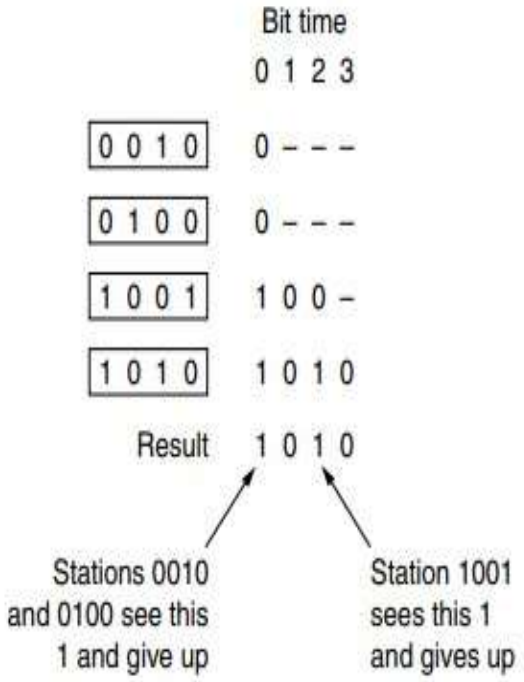
- Note that we do not need a physical ring to implement token passing.
- The channel connecting the stations might instead be a single long bus.
- Each station then uses the bus to send the token to the next station in the predefined sequence.
- Possession of the token allows a station to use the bus to send one frame, as before. This protocol is called token bus.

Performance of token passing

- The performance of token passing is similar to that of the bit-map protocol, though the contention slots and frames of one cycle are now intermingled.
- After sending a frame, each station must wait for all N stations (including itself) to send the token to their neighbors and the other N − 1 stations to send a frame, if they have one.
- A subtle difference is that, since all positions in the cycle are equivalent, there is no bias for low- or high-numbered stations.
- For token ring, each station is also sending the token only as far as its neighboring station before the protocol takes the next step.
- Each token does not need to propagate to all stations before the protocol advances to the next step.
- Token rings have cropped up as MAC protocols with some consistency.
- An early token ring protocol (called **‘‘Token Ring’’ and standardized as IEEE 802.5**) was popular in the 1980s as an alternative to classic Ethernet.
- In the 1990s, a much faster token ring called **FDDI (Fiber Distributed Data Interface)** was beaten out by switched Ethernet.
- In the 2000s, a token ring called **RPR (Resilient Packet Ring) was defined as IEEE 802.17** to standardize the mix of metropolitan area rings in use by ISPs

Binary Countdown

- A problem with the basic bit-map protocol, and by extension token passing, is that the overhead is 1 bit per station, so it does not scale well to networks with thousands of stations
- binary countdown
- A station wanting to use the channel now broadcasts its address as a binary bit string, starting with the high order bit.
- All addresses are assumed to be the same length.
- The bits in each address position from different stations are BOOLEAN ORed together by the channel when they are sent at the same time.
- It was used in Datakit (Fraser, 1987).
- It implicitly assumes that the transmission delays are negligible so that all stations see asserted bits essentially instantaneously.



Efficiency

- The channel efficiency of this method is d /(d + log2 N).
- If, however, the frame format has been cleverly chosen so that the sender's address is the first field in the frame, even these log2 N bits are not wasted, and the efficiency is 100%.
- Binary countdown is an example of a simple, elegant, and efficient protocol